



Department of Electrical and Electronics Engineering
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE3403 Measurements and Instrumentation

Name of the Faculty member: Mrs. S. Sharmila Kumari, AP/EEE

Innovative Practice Description

- Unit / Topic: Unit I / Errors in measurement
- Course Outcome: CO 1
- Topic Learning Outcome: TLO 4
- Activity Chosen: One minute paper
- Justification:

- Explain various types of errors

• The topic Errors in Measurements has 9 types of errors in instruments. Since each type has its own definition, limitation and examples. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of errors which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 63

Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of errors in measurements concept for 3 minutes.
- ✓ Then I told them to write as much as they can within a short period of time (1 minute)
- ✓ Finally, I collected the papers from each column (1 minute)

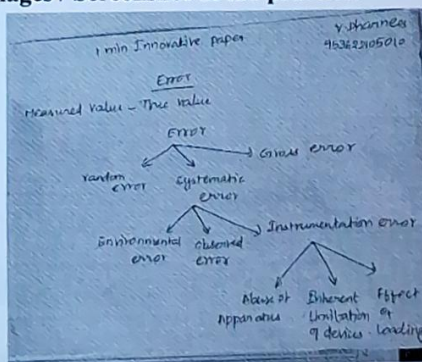
• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO1	PSO3
C213.1	3	1	1	1	1	1	2	1

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO10	PO12	PSO1	PSO3
	3	1	1	1	1	1	2	1
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the basics of measurement and instruments.	Students will identify the mathematical, engineering and other relevant knowledge that applies to an Analog and digital measurement systems.	Students will determine the design objectives and functional requirements of the instruments.	Students will examine the relevant methods, tools and techniques of system calibration.	Students will be able to create flow in a document or presentation.	Students will describe the rationale for the requirement for continuing professional development.	Students able to analyze the performance of Electrical parameter measuring instruments.	Students able to interpret the working of measuring instruments used in Industrial automation sector.

• Images / Screenshot of the practice:



❖ *Reflective Critique:*

1. **Pre-implementation Reflection :**

• **Benefits:**

- Students can able to attend the question even in the questions are in indirect form.

- Students can able to explain the concepts in examination without any confusion.

- **Challenges:**

- In the class mostly boys hesitate to answer the questions.
- Time utilization for conducting activity.

2. Post-implementation Reflection :

- **Benefits:**

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- **Challenges:**

- Slow learners were not able to understand some topics during discussion hours.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain various types of errors present in measurements.
- ✓ The success of the activity was evaluated by asking the same question in

Internal Assessment test I – Around 60% of students answered.

References:

- ❖ <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-one-minute-paper>

S. S.
20/3/24

Signature of Faculty Member

S. K.
20/03/24
HOD



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Department of Electrical and Electronics Engineering Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE3403 Measurements and Instrumentation

Name of the Faculty member: Mrs. S. Sharmila Kumari, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit II / Principle and types of analog voltmeters and ammeters
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 6
- **Activity Chosen:** Demonstration
- **Justification:**
 - Explain the principle and types of analog meter
 - The topic Principle and types of analog meter has 7 types of instruments. Since each type has its own principle of operation, torque equation and construction. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of meters which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 64,

Reporter: Myself

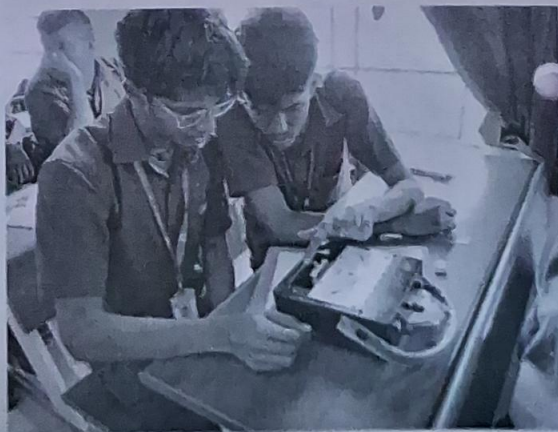
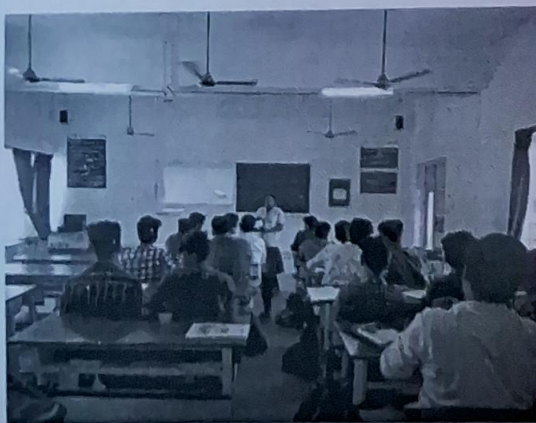
- At the end the Class (Last 5 minutes)
 - I asked the students to think about various types of meters in measurements concept for 2 minutes.
 - Then I told them to Pair with their neighbours and discuss about the construction parts of instruments for another 1 minute.
 - Finally, I show the PMMC and PMMI ammeter for each row (3 minutes)
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO1	PSO3
C213.2	3	2	1	1	1	1	2	1

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO10	PO12	PSO1	PSO3
	3	1	1	1	1	1	2	1
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the basics of measurement and instruments.	Students will identify the mathematical, engineering and other relevant knowledge that applies to an Analog and digital measurement systems.	Students will determine the design objectives and functional requirements of the instruments.	Students will examine the relevant methods, tools and techniques of system calibration.	Students will be able to create flow in a document or presentation.	Students will describe the rationale for the requirement for continuing professional development.	Students able to analyze the performance of Electrical parameter measuring instruments.	Students able to interpret the working of measuring instruments used in Industrial automation sector.

• Images / Screenshot of the practice:



❖ *Reflective Critique:*

1. **Pre-implementation Reflection:**

• **Benefits:**

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

• **Challenges:**

- In the class mostly boys hesitate to answer the questions.
- Time utilization for conducting activity.

2. **Post-implementation Reflection:**

• **Benefits:**

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

• **Challenges:**

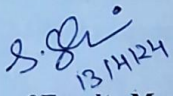
- Slow learners were not able to understand some topics during discussion hours.

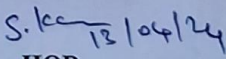
❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students can able to explain the Analog meters (MC, MI, and Electrodynamometer) construction and working principle.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 80% of students answered.**

References:

1. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2010.
2. <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-demonstrations>


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Department of Electrical and Electronics Engineering Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE3403 Measurements and Instrumentation

Name of the Faculty member: Mrs. S. Sharmila Kumari, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit III / AC Bridges
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO12
- **Activity Chosen:** Demonstration
- **Justification:**

There are totally 5 types of AC Bridge circuit and 2 types of DC bridge circuits given in the Course. Students find quite difficult to remember all those bridge circuits without revision. Rather than the traditional way of repeating the concepts in the class, this activity helped the students to revise all types of bridge circuit effectively.

- **Time Allotted for the Activity:** 5 minutes
- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 64,

Reporter: Myself

- At the end the Class (Last 5 minutes)
- I asked the students to think about various types of meters in measurements concept for 2 minutes.
- Then I told them to Pair with their neighbours and discuss about the construction parts of instruments for another 1 minute.
- Finally, I show the Schering bridge kit for each row (3 minutes)

CO – PO / PSO mapping:

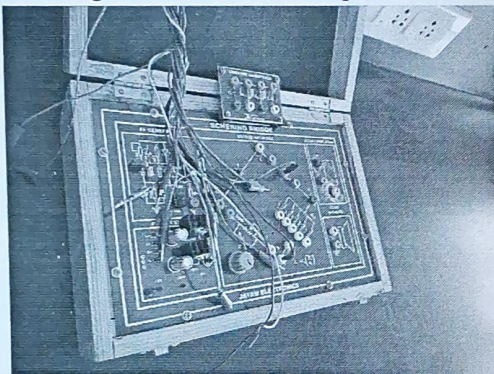
CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO3
C213.3	3	2	1	1	1	1	1	2	1

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO3
	3	1	1	1	1	1	1	2	1
Justification for correlation	Students will apply the knowledge of Electrical	Students will identify the mathematical, engineering	Students will determine the design	Students will examine the	Students will demonstrate effective communication.	Students will be able to create flow in a	Students will describe the rationale for the requirement for	Students able to analyze the performance of Electrical parameter	Students able to interpret the working of measuring

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO3
	3	1	1	1	1	1	1	2	1
	and Electronics engineering fundamental concepts to understand the basics of measurement and instruments.	and other relevant knowledge that applies to an Analog and digital measurement systems.	objectives and functional requirements of the instruments.	relevant methods, tools and techniques of system calibration.	problem-solving, conflict resolution and leadership skills with their team through Active Learning Methods.	document or presentation.	continuing professional development.	measuring instruments.	instruments used in Industrial automation sector.

• Images / Screenshot of the practice:



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The students enjoyed the activity. After the class many students requested me to conduct more such activities in future.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain the various types of AC Bridges.
- ✓ The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 60% of students answered.

• References:

1. <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-demonstrations>

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18/11/24
Signature of Faculty Member

S. K. S.
18/04/24
HOD



Department of Electrical and Electronics Engineering

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE3403 & Measurements & Instrumentation

Name of the Faculty member (s): Mrs.S.Sharmila Kumari

Innovative Practice Description

- **Unit / Topic:** Unit IV / Smart Sensors
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO18
- **Activity Chosen:** Think pair share
- **Justification:**
 - ✓ Since it is a new topic, the activity will make the students explore about the applications of technologies they have already learned.
 - ✓ It enhances the team skills of students
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

The sensors used to measure temperature, pressure, flow, angular velocity were taught in the earlier classes. In this class an introduction about the term “smart” has been given. Then I have posted a question to the students What makes normal sensor, a smart sensor?. Time has been given to students to think and jot down their ideas after discussing with their peers. After discussion, one person from a team presented the requirements of a smart sensor.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	2	1	-	-	-	-	-	1	1	-	1	2	-	1

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution	
Principle of operation of Synchronous motor	
– Think Pair Share	
	

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students stated that they understood the concept, and the block diagram given in the reading material is just indicative.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The activity enhanced the participative learning capability of students

- ❖ **Challenges faced in implementation:**

- ✓ The peer discussion deviated from the topic occasionally, so the student needed to be regulated often to create the interest

References:

1. <https://spedellreadingstrategies.weebly.com/think-pair-share.html>
2. <https://www.techtarget.com/iotagenda/definition/smart-sensor>

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Department of Electrical and Electronics Engineering

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: B.E, IV Semester & EEE

Course Code & Title: EE3403 & Measurements & Instrumentation

Name of the Faculty member: Dr.P.Prem, ASP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit V / Introduction to Virtual Instrumentation
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO24
- **Activity Chosen:** Flipped Classroom
- **Justification:**

This topic is chosen, since it can be easily understood by students. Students can improve their understanding on this topic by this activity. More time for discussion is given the class unlike traditional learning. This activity greatly helps the student to know the concept much deeper.

- **Time Allotted for the Activity:** 25 Minutes

- **Details of the Implementation:**

- Each team was asked to conduct a literature survey on the assigned topic. They were required to include at least 5 references related to the topic, using the materials already shared. Students within each team were tasked with creating a PowerPoint presentation to present their topic. After the presentation, students could ask questions to the presenting team.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO9	PO10	PSO1
CO1	3	2	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO3	PO9	PO10	PSO1
	3	2	1	1	2
Justification for correlation	Students will apply the knowledge of Electrical and Electronics engineering fundamental concepts to understand the basics concept of digital instrumentation	Students will determine the design objectives and functional requirement of the instruments and arrive at specification	Students function effectively as team and as an individual in this activity	Students will be able to effectively communicate their ideas in this activity	Students able to analyze the performance of Electrical instruments by measured parameters

- **Images / Screenshot of the practice:**

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

The students found the activity engaging. They came up with new ideas to explain the concept of virtual instrumentation, one of the methods has been used in the learning material uploaded in CANVAS

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students related the concepts learned to real world applications like SCADA based data acquisition systems.

- ❖ ***Challenges faced in implementation:***

Few students were found non participative. So, they were permitted to communicate in native language to improve their confidence

References:

- ❖ <https://tophat.com/glossary/f/flipped-classroom/#:~:text=A%20flipped%20classroom%20refers%20to,with%20their%20peers%20or%20instructor.>
- ❖ <https://www.ni.com/en/shop/labview/virtual-instrumentation.html>

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